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General Comment

The AI industry is built on the mass laundering of behavior that would be clear copyright infringement if done by a human (e.g., near-verbatim replication of pages of copyright software). Many of the behaviors of deployed systems would be illegal if performed by a human (violating civil rights law, medical and legal licensure laws, employment law, etc.). The most crucial innovation of this industry is to exploit ambiguities and loopholes in existing law to create systems that allegedly evade liability. The products would never pass safety or fitness for purpose standards that other industries are expected to comply with.

https://spectrum.ieee.org/ai-code-generation-ownership
https://doi.org/10.1145/3531146.3533158
https://doi.org/10.1145/3442188.3445922

And yet these AI companies have been an investor's dream speculation instrument because of the promise of mass exploitation of a captive populace. Having reached market saturation and being over-extended with respect to copyright law, these companies are increasingly desperate for government assistance to justify their extreme valuations. (None of these companies have a profitable product and their P/E ratios are huge. They will need to spike revenue soon, but the product isn't good enough for people to pay that premium voluntarily.) The government assistance they seek will come at the cost of constitutional rights and global human rights, as well as severe global environmental harm.

It's important to realize that AI does not have a meaningful technical definition. NAIRR (https://www.ai.gov/wp-content/uploads/2023/01/NAIRR-TF-Final-Report-2023.pdf) attempted such a technical definition, yet it clearly applies to all operational numerical weather prediction since their dawn in the 1950s. And yet AI practitioners will invent additional criteria to exclude such "classical" techniques. Indeed, there is a spectrum between mechanism-first and data-first systems, with often-fruitful synergies between them.

While bearing negligible technical information, use of the name "AI" is a cultural affinity signal of disregard for harms caused by shoddy systems. Any future AI initiative should engage deeply with how the field is repeating the crisis confronted by the computational fluids community in the 1980s (see attached editorial board statement for the Journal of Fluids Engineering). The aerospace industry and research community of the 1980s were forced to clean up their own house because they were not awash with hundreds of billions in investment dollars with government standing by to rescue them from consequences for producing unreliable and poorly-performing aircraft. Rigorous Verification and Validation practices emerged from the 1980s crisis in fluids engineering, and have served well to improve efficiency and capability while avoiding the massive engineering catastrophes of earlier eras.

The AI industry needs to stop dodging liability and reckon with a mandate to ensure their products are safe and reliable, fit for purpose, and able to pass strict regulatory scrutiny, much like the aerospace industry. If AI companies wish to establish that training is not copyright infringement, they should adopt practices capable of establishing that with human agents, namely clean-room design (https://en.wikipedia.org/wiki/Clean-room_design). Today's AI systems are incapable of being trained by clean-room design, but doing so would be a real advance. If a human doing a task would be liable, the operators of an AI system assigned that task should assume the same direct liability. If a task would be illegal if performed by a human, it must also be illegal to operate an AI system performing that task. Insisting on such criteria will create many research opportunities, and we will find that many previously-claimed results (as well as industry practices) do not stand up to scrutiny. NSF should focus convergent research attention on the design of effective regulation and standards.

Attachments

RoacheGhiaWhite-JFEEditorialPolicyStatementControlOfNumericalAccuracy-1986

Journal of Fluids Engineering

Editorial

Editorial Policy Statement on the Control of Numerical Accuracy

A professional problem exists in the computational fluid dynamics community and also in the broader area of computational physics. Namely, there is a need for higher standards on the control of numerical accuracy.

The numerical fluid dynamics community is aware of this problem but, although individual researchers strive to control accuracy, the issue has not to our knowledge been addressed collectively and formally by any professional society of journal editorial board. The problem is certainly not unique to the JFE and came into even sharper focus at the 1980-81 AFOSRHTTM-Stanford Conference on Complex Turbulent Flows. It was a conclusion of that conference's Evaluation Committee[1] that, in most of the submissions to that conference, it was impossible to evaluate and compare the accuracy of different turbulence models, since one could not distinguish physical modeling eerors from numerical errors related to the algorithm and grid. This is especially the case for first-order accurate methods and hybrid methods.

The practice of publishing comparisons based on coarse grid solutions, without systematic truncation error testing, may have been acceptable in the past. Certainly ten to fifteen years ago any calculation was of interest, and much of the exploratory work deserved publication, as many researchers lacked the computational power or funds to do a thorough and systematic error estimation. We are of the opinion that this practice, however understandable in the past, is outmoded and that, with powerful computers becoming more common, standards should be raised. Consequently, this journal hereby announces the following policy:

> *The Journal of Fluids Engineering will not accept for publication any paper reporting the numerical solution of a fluids engineering problem that fails to address the task of systematic truncation error testing and accuracy estimation.*

Although the formal announcement of this journal policy is new, it has been the practice of many of our conscientious reviewers. Thus the present announcement is not a change in policy so much as a clarification and standardization.

Methods are available to accomplish this task, such as Richardson extrapolation (when applicable), calculations with a high- and low-order method on the same grid, and straightforward repeat calculations with finer or coarser grids. As in the case of experimental uncertainty analysis, ". . . any appropriate analysis is far better than none as long as the procedure is explained."[2] Whatever the authors use will be considered in the review process, but we must make it clear that *a single calculation in a fixed grid will not be acceptable*, since it is impossible to infer an accuracy estimate from such a calculation. Also, the editors will not consider a reasonable agreement with experimental data to be sufficient proof of accuracy, especially if any adjustable parameters are involved, as in turbulence modeling.

We recognize that it can be costly to do a thorough study, and that many practical engineering calculations will continue to be performed on a single fixed grid. However, this practice is insufficient for publication in an archival journal.

[1]Emmons, H. W. (Chairman), "Evaluation Committee Report," pp. 979-986 in *Proc. 1980-81 AFOSR-HTTM-Stanford Conference on Complex Turbulent Flows*, Vol. II, "Taxonomies, Reporter's Summaries, Evaluation, and Conclusions," Thermosciences Division, Mechanical Engineering Department, Stanford University.

[2]Kline, S. J ., "The Purposes of Uncertainty Analysis," ASME Journal of Fluids Engineering, Vol. 107, June 1985, pp. 153-164.

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